

SKILLS

- **AI / Deep Learning:** PyTorch, TensorFlow, NumPy, HuggingFace, neural-network inference, model evaluation, quantization concepts
- **Performance Engineering:** profiling, runtime analysis, latency SLOs, parallelism, caching, resource optimization, production debugging
- **Programming:** Python, Go, TypeScript, JavaScript, SQL, REST APIs, distributed systems, algorithms, data structures
- **ML Systems:** inference pipelines, batch/online scoring, model validation, drift detection, anomaly detection, reproducible experiments
- **Linux / Cloud:** Linux, Docker, Kubernetes, AWS EKS, GCP, Terraform, CI/CD, microservices, secure deployment
- **Data / Observability:** OpenTelemetry, Kafka, Vector, Fluent Bit, Logstash, ELK, distributed tracing, Prometheus
- **Leadership:** technical roadmaps, cross-functional delivery, architecture reviews, code reviews, mentoring, stakeholder communication

EXPERIENCE

- **Royal Bank of Canada (RBC)** Toronto, Canada
Staff Software Developer & Enterprise Architect Nov 2022 – Present
 - **High-Performance Telemetry Stack:** Architected OpenTelemetry infrastructure using Vector, Fluent Bit, Kafka, and Logstash that ingests 150 TB/day across 10,000+ systems; optimized resource usage, schema contracts, and vendor-neutral data paths for reliable low-latency processing.
 - **Profiling, Debugging & Runtime Visibility:** Standardized telemetry and diagnostic workflows adopted by 20+ engineering teams, enabling performance profiling, distributed tracing, latency SLOs, anomaly detection, and rapid root-cause analysis across production workloads.
 - **ASB Assist – Neural AI Feature (In Development):** Building an LLM + RAG capability with retrieval and inference pipelines, offline evaluation, regression testing, grounding controls, audit trails, and production-readiness gates for regulated enterprise use cases.
 - **0-to-1 Software Platform Ownership:** Led production delivery of a patent-pending Node.js/TypeScript/React platform; wrote 99.6% of production code and 100% of core logic for a rules-based recommendation engine, real-time rendering, service matching, and auto-certification.
 - **Cross-Functional Technical Leadership:** Lead 5 engineers through design, implementation, code reviews, ADRs, testing, release readiness, and operational handoff while coordinating product, risk, security, platform, and architecture stakeholders.
- **HealthCard (Acquired Startup)** Remote
Blockchain Architect & First Full Stack Engineer Aug 2021 – Jul 2022
 - **Production Neural-Network Inference:** Integrated transformer-based document classification into medical and insurance workflows; deployed inference APIs on AWS EKS with monitoring, audit logging, and feedback loops, reducing manual review by 60%.
 - **Full-Stack Platform Engineering:** Built the product from scratch across React/TypeScript, Node.js services, verification logic, and containerized Linux workloads, balancing latency, security, resource usage, and commercial-quality releases.
- **Dalhousie University** Halifax, Canada
Research & Teaching Assistant Jan 2021 – Aug 2022
 - **Compiler / Runtime Algorithms:** Researched call-stack decision algorithms and built Python tracing tools for compiler/runtime analysis; taught Software Development & Algorithms and graduate Advanced Cloud Computing.
- **Canada Revenue Agency (CRA)** Ottawa, Canada (Remote)
IT Developer / Software Developer Co-op Apr 2022 – Nov 2022
 - **Commercial-Quality Software Delivery:** Modernized a federal tax platform serving millions of Canadians and established automated SonarQube quality gates for maintainability, accessibility, and secure releases.

PROJECTS

- **Edge Neural-Network Performance Lab:** Built a Python, PyTorch, TensorFlow, and NumPy benchmarking prototype to compare model latency, throughput, memory use, numerical accuracy, and quantization tradeoffs across constrained inference configurations.
- **Flight Price Optimization Agent:** Developed a DQN/DDQN reinforcement-learning agent for dynamic pricing; modelled demand curves on historical data and improved simulated outcomes by 18% over baseline.

EDUCATION

- **Dalhousie University** Halifax, Canada
Master of Applied Computer Science, GPA: 3.9/4.0 Jan 2021 – Aug 2022
- **University of Toronto** Toronto, Canada
Enterprise Architecture Certification (3-semester program) Completed 2024
- **Gujarat Technological University** Ahmedabad, India
B.E. Computer Engineering, GPA: 3.92/4.0 Aug 2016 – Aug 2020